THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The LCM of 30, 42 and 66 is: 1

(a) 1155 **(b)** 6930 **(c)** 2310 **(d)** 1260

2. The least number that is divisible by every number from 1 to 6 is: 1

(a) 60 **(b)** 360 **(c)** 120 **(d)** 30

3. The largest number dividing 67, 103 and 175 leaving remainder 4 in each case is: 1

(a) 18 **(b)** 9 **(c)** 1 **(d)** 3

4. The smallest positive integer by which 12 must be multiplied to obtain a perfect square is: 1

(a) 6 **(b)** 3 **(c)** 12 **(d)** 2

5. Which of these is a rational number? 1

(a) $2\sqrt{5}$ **(b)** $\frac{\sqrt{12}}{\sqrt{3}}$ **(c)** $\sqrt{2}+\sqrt{3}$ **(d)** $\sqrt{4}$

6. A number ends with exactly 2 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^2 \times 3$ **(b)** $3^2 \times 7$ **(c)** $2^3 \times 5^1 \times 3$ **(d)** $2^1 \times 5^2 \times 3$

7. The LCM of 36, 48 and 72 is: 1

(a) 5184 **(b)** 1728 **(c)** 72 **(d)** 144

8. The least number that is divisible by every number from 1 to 7 is: 1

(a) 840 **(b)** 2520 **(c)** 210 **(d)** 420

9. **Assertion (A):** The HCF of 12 and 18 is 6 and their LCM is 36. 1

Reason (R): For two positive integers a and b, HCF(a, b) x LCM(a, b) = a x b.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The number $5-2\sqrt{3}$ is irrational. 1

Reason (R): The sum of a rational number and an irrational number is irrational.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two numbers have HCF 12 and are in the ratio $3 : 5$. Find the numbers and their LCM. 2

12. Find the smallest positive integer that leaves remainders 7 and 13 when divided by 12 and 18 respectively. 2

13. Prove that $3\sqrt{7}$ is irrational, given that $\sqrt{7}$ is irrational. 2

14. Two numbers have HCF 15 and are in the ratio $4 : 7$. Find the numbers and their LCM. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A, B and C take 18, 24 and 36 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

16. Find the greatest number that divides 1251 and 9377 leaving remainder 1 in each case. 3

17. Is $5 \times 6 \times 7 + 7$ a composite number? Justify. 3

OR

A, B and C take 20, 28 and 35 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

18. Find the greatest number that divides 1157 and 1397 leaving remainder 5 in each case. 3

OR

Is $6 \times 7 \times 8 + 8$ a composite number? Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school is arranging chairs for the annual day. There are 144 chairs in the front block and 180 chairs in the back block. The teacher wants to arrange each block in rows so that every row has the same number of chairs, and every row in both blocks has equal length. 5

(i) Express 144 and 180 as products of prime factors.

(ii) Find the greatest possible number of chairs in each row.

(iii) How many rows will there be in all?

(iv) The coordinator proposes one more block of 216 chairs with the same rule. Find the greatest row length that works for all three blocks and the total number of rows.

20. Three sets of English, Hindi and Mathematics books have 144, 192 and 288 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

Prove that $\frac{1}{\sqrt{3}}$ is irrational and hence that $\frac{5}{\sqrt{3}}$ is irrational. 5

END OF QUESTION PAPER — Total: 40 Marks